

# L7: Hierarchical Clustering Approaches – Group Similarity

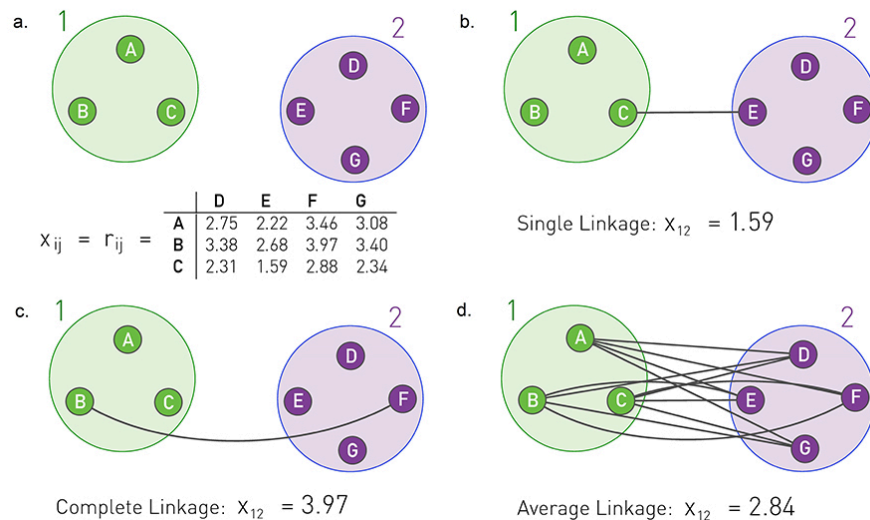


Image 9.10 from networksciencebook.com Three approaches, called single, complete and average cluster similarity, are frequently used to calculate the community similarity from the node-similarity matrix  $x_{ij}$

How to compute the similarity between two groups of nodes (as opposed to individual nodes)?

In other words, if we are given two groups of nodes, say 1 and 2, and we know the pairwise node similarities, how to compute the similarity between groups 1 and 2?

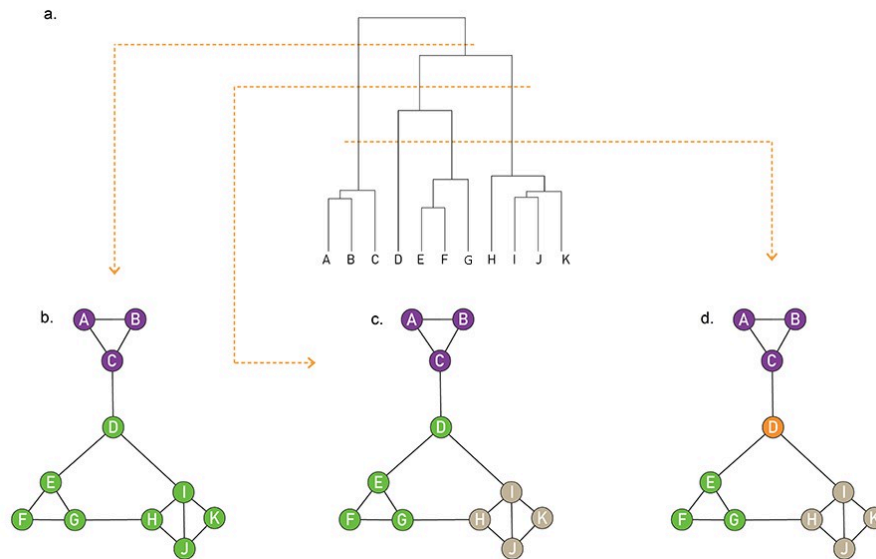
**There are three ways to do so:**

1. *Single linkage*: the similarity of groups 1 and 2 is determined by the minimum distance (i.e., maximum similarity) across all pairs of nodes in groups 1 and 2.
2. *Complete linkage*: the similarity of groups 1 and 2 is determined by the maximum distance (i.e., minimum similarity) across all pairs of nodes in groups 1 and 2.
3. *Average linkage*: the similarity of groups 1 and 2 is determined by the average distance (i.e., average similarity) across all pairs of nodes in groups 1 and 2.

The visualization illustrates the three approaches. Note that this figure gives the pairwise distance between nodes. The similarity between two nodes can be thought of as inversely related to their distance.

Average linkage is the most commonly used metric.

## Where to “cut” the dendrogram and computational complexity



A hierarchical tree in which any cut of the hierarchical tree offers a potentially valid partition (Image 9.15) from networksciencebook.com

Now that we have defined a similarity metric for two nodes (*based on the number of common neighbors*), and we have also learned how to calculate a similarity value for two groups of nodes, we can design the following iterative algorithm to compute a hierarchical clustering dendrogram in a bottom-up manner.

We start with each node represented as a leaf of the dendrogram. We also compute the matrix of  $N^2$  pairwise node similarities.

In each step, we select the two nodes, or two groups of nodes more generally, that has the highest similarity value – and merge them into a new group of nodes. This new group forms a larger community and the corresponding merging operation corresponds to a new branching point in the dendrogram.

The process completes until all the nodes belong in the same group (*root of the dendrogram*).

Note that depending on where we “*cut*” the dendrogram we will end up with a different set of communities. For instance, the lowest horizontal yellow line at the visualization corresponds to four communities (*green, purple, orange, and brown*) – note that node *D* forms a community by itself.

The computational complexity of this algorithm, which is known as Ravasz algorithm, is  $O(N^2)$  because the algorithm requires:

1.  $O(N^2)$  for the initial calculation of the pairwise node similarities
2.  $O(N)$  for updating the similarity between the newly formed group of nodes and all other groups – and this is performed in each iteration, resulting in  $O(N^2)$
3.  $O(N \log N)$  for constructing the dendrogram

## Food For Thought

Try to think of another agglomerative hierarchical clustering approach, based on a different similarity metric.